

C++ Program (Stack using Array)

```
#include<iostream>

using namespace std;

#define MAX 5

class Stack
{
    int stack[MAX];
    int top;

public:

    Stack()
    {
        top=-1;
    }

    void push()
    {
        if(top==MAX-1)
        {
            cout<<"Stack Overflow"<<endl;
        }
        else
        {
            int parcel;
            cout<<"Enter Parcel ID : ";
            cin>>parcel;
```

```
        top++;
        stack[top]=parcel;

        cout<<"Parcel Added Successfully"<<endl;
    }
}

void pop()
{
    if(top==-1)
    {
        cout<<"Stack Underflow"<<endl;
    }
    else
    {
        cout<<"Delivered Parcel ID : "<<stack[top]<<endl;
        top--;
    }
}

void display()
{
    if(top==-1)
    {
        cout<<"No Parcels"<<endl;
    }
    else
    {
        cout<<"Parcels in Stack"<<endl;

        for(int i=top;i>=0;i--)
        {
```

```
        cout<<stack[i]<<endl;
    }
}
};
```

```
int main()
{
    Stack s;
    int choice;

    do
    {
        cout<<"\n1.Push Parcel";
        cout<<"\n2.Pop Parcel";
        cout<<"\n3.Display";
        cout<<"\n4.Exit";

        cout<<"\nEnter Choice : ";
        cin>>choice;

        switch(choice)
        {
            case 1:
                s.push();
                break;

            case 2:
                s.pop();
                break;

            case 3:
```

```
s.display();  
  
break;  
  
case 4:  
    cout<<"Program End";  
    break;  
  
default:  
    cout<<"Invalid Choice";  
}  
  
}while(choice!=4);  
  
return 0;  
}
```

OUTPUT:

- 1.Push Parcel
- 2.Pop Parcel
- 3.Display
- 4.Exit

Enter Choice : 1

Enter Parcel ID : 101

Parcel Added Successfully

Enter Choice : 1

Enter Parcel ID : 102

Parcel Added Successfully

Enter Choice : 3

Parcels in Stack

102

101

Enter Choice : 2

Delivered Parcel ID : 102

Enter Choice : 3

101